chinesepowered / hack-commons

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frontend	refactor	1 hour ago
solana	checkpoint	4 hours ago
.gitattributes	Initial commit	6 hours ago
.gitignore	checkpoint	4 hours ago
README.md	refactor	1 hour ago
SUBMISSION.md	fixes	1 hour ago
deploy.md	refactor	1 hour ago
docs-reference.md	checkpoint	4 hours ago
hackathon.md	plan	5 hours ago
plan.md	Update plan.md	5 hours ago

README

🤖 AgentCommerce — Multi-Agent Economy on Solana

> AI agents that discover, negotiate, and pay each other on-chain — governed by verified humans.

What if AI agents could form their own economy? AgentCommerce is an autonomous agent marketplace where specialized AI agents register on-chain identities via **Metaplex**, discover each other's capabilities, competitively bid on tasks, and settle payments using the **x402 protocol** on Solana — all in real-time, all on-chain, all transparent.

Humans stay sovereign. They post tasks, set budgets, and approve high-value actions. Agents do the work.

🎬 Demo Video

[Link to 4-min demo video]

⚡ How It Works

```
👤 Human posts task
   → 🧠 Orchestrator decomposes into sub-tasks
   → 💰 Agents bid competitively
   → ✅ Best agent wins each sub-task
   → ⚙️ Work gets done (research, analysis, execution)
   → 💸 Payment settles on Solana via x402
   → 📊 Results returned to human
```

Everything happens live on the dashboard — watch agents think, negotiate, and pay each other in real-time.

🏗️ The Agent Network

Agent	Role	What it does
🧠 **Orchestrator**	Coordinator	Decomposes tasks, collects bids, assigns work, manages x402 payments
🔍 **Researcher**	Intelligence	Gathers web data via Unbrowse, synthesizes market findings
📊 **Analyst**	Processing	Risk assessment, yield analysis, actionable recommendations
⚡ **Executor**	On-chain	Swaps, LP positions, staking — human approval for high-value txns
🏢 **Frontier Tower**	Services	Room booking, bounty posting, expert matching for the 16-floor SF innovation hub

🔑 Key Features

- 🪪 **On-chain Agent Identity** — Each agent registered on Metaplex Agent Registry with discoverable profiles and verifiable credentials
- 💸 **x402 Payments** — HTTP-native micropayments between agents. Agent calls service → gets 402 → pays on Solana → gets result. No middleman.
- 📡 **Real-time Dashboard** — Watch agents negotiate, bid, and transact live with clickable Solana explorer links
- 🛡️ **Human Sovereignty** — Verified humans (via Human Passport) control task posting and approve high-value actions
- 🎙️ **Voice Interface** — Talk to the orchestrator via ElevenLabs ("What are you working on?") with text fallback always available
- 🔌 **Pluggable Architecture** — Every integration can be independently enabled/disabled. Core works with zero API keys.

🏛️ Architecture

```
┌──────────────────────────────────────────────────┐
│        📊 Next.js (Dashboard + API Routes)        │
│   Agent cards · Live activity · Txn feed · Chat/Voice │
│   Task routing · Agent orchestration · x402 protocol  │
└──────────────────────────────────────────────────┘
   │        │        │        │        │
   ▼        ▼        ▼        ▼        ▼
  🧠       🔍       📊       ⚡       🏢
 Orch.    Res.    Analyst   Exec.   Frontier
 Agent    Agent    Agent    Agent    Tower
   │        │        │        │        │
   ▼        ▼        ▼        ▼        ▼
┌──────────────────────────────────────────────────┐
```

```
│     Solana Devnet · Metaplex Agent Registry · x402     │
└────────────────────────────────────────────────────────┘
```

💸 x402 Payment Flow

This is how agents pay each other — pure HTTP, no intermediaries:

```
🤖 Agent A calls Agent B's service endpoint
        │
        ▼
🔒 Agent B returns HTTP 402 + PaymentRequirements
   (recipient wallet, amount in SOL, network)
        │
        ▼
✍️ Agent A creates + signs SOL transfer on Solana
        │
        ▼
📨 Agent A retries with X-Payment header (base64 tx)
        │
        ▼
✅ Agent B verifies on-chain → executes service → returns result
```

🧰 Tech Stack

Layer	Technology
App	Next.js, React, TypeScript, Tailwind CSS, pnpm
Blockchain	Solana (devnet), @solana/web3.js, Metaplex Agent Registry, x402
LLM	Any OpenAI-compatible API (OpenAI, Together, Groq, etc.) / mock fallback
Web Intelligence	Unbrowse (fallback: standard HTTP)
Voice	ElevenLabs (fallback: text chat)
Auth	Human Passport (fallback: wallet auth)

🚀 Quick Start

Prerequisites

- Node.js 18+ with pnpm

Run

```
cd frontend
cp .env.example .env.local  # Add API keys as needed (works without any!)
pnpm install
pnpm dev
```

Dashboard will be at http://localhost:3000

Register Agents on Solana

```
cd solana
pnpm install
pnpm run register
```

Initialize Wallets

Click "**Initialize Wallets**" on the dashboard, or:

```
curl -X POST http://localhost:3000/api/wallets/init
```

📡 API

Core

Method	Path	Description
GET	/api/health	Health check
GET	/api/agents	List all agents with status and balances
POST	/api/tasks	Submit a task to the agent economy
GET	/api/events	Poll for real-time agent activity

Wallets & Payments

Method	Path	Description
POST	/api/wallets/init	Initialize agent wallets + airdrop devnet SOL
GET	/api/wallets	List wallets with balances
POST	/api/x402/service/{agent_id}	x402-protected agent service endpoint
GET	/api/x402/prices	List per-agent service prices
POST	/api/demo/x402-flow	Demo full x402 payment flow

Chat & Voice

Method	Path	Description
POST	/api/chat	Chat with orchestrator (text)
POST	/api/chat/voice	Chat with voice response
GET	/api/voice/status	Check voice availability

🔌 Pluggable Integrations

Every integration gracefully degrades. The entire system works with **zero API keys** — just `pnpm install` and go.

Integration	Purpose	Fallback
Unbrowse	Web data extraction for Researcher	Mock research data

Integration	Purpose	Fallback
ElevenLabs	Voice interface (talk to your agents!)	Text-only chat
Human Passport	Sybil-resistant auth for task posting	Basic wallet auth
LLM API	Any OpenAI-compatible endpoint (OpenAI, Together, Groq, etc.)	Realistic mock responses

🛡️ Human Sovereignty

AgentCommerce embodies **Voluntary Accountability**. AI agents act autonomously within human-defined boundaries:

- ✅ Only verified humans can post tasks
- ✅ Humans set budgets and approval thresholds
- ✅ High-value transactions require human sign-off
- ✅ Every agent action is transparent on the real-time dashboard
- ✅ Agents explain their reasoning at every step

Agents do the work. Humans hold the keys.

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

chinesepowered

Languages

Python 61.8% TypeScript 36.3% CSS 1.3% JavaScript 0.6%

Suggested workflows

Based on your tech stack

Python application Configure

Create and test a Python application.

Django Configure

Build and Test a Django Project

Pylint Configure

Lint a Python application with pylint.

More workflows

Dismiss suggestions